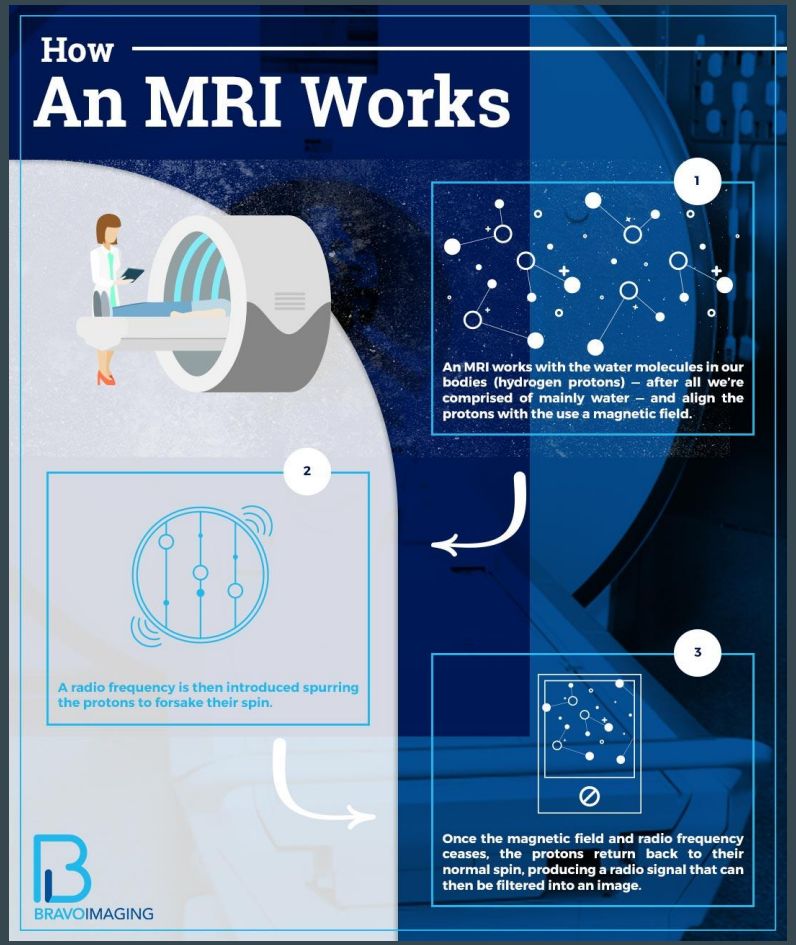
Lightning Presentations: MRI Reconstruction

Problem Statement & Motivation

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Shorter scan times = more accessible treatment



Technical Approach

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

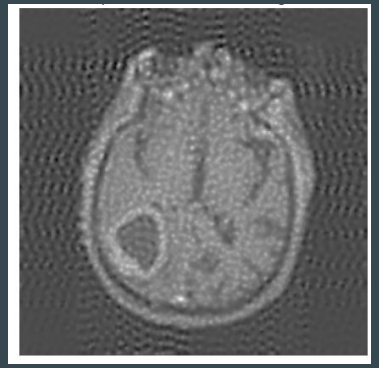
There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We start with **Gaussian and LaPlacian kernels** .

Initial Results pt. 1

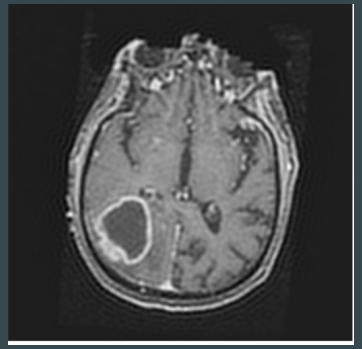
Metric	*LaPlacian Kernels*	*Gaussian Kernels*	*Zero-Filled FFT*
MSE	0.0186	0.00529	0.010049
PSNR	17.30 dB	22.78 dB	19.98 dB

Original Image

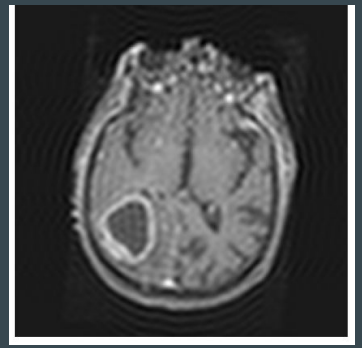
Laplace Reconstruction



Gaussian Reconstruction



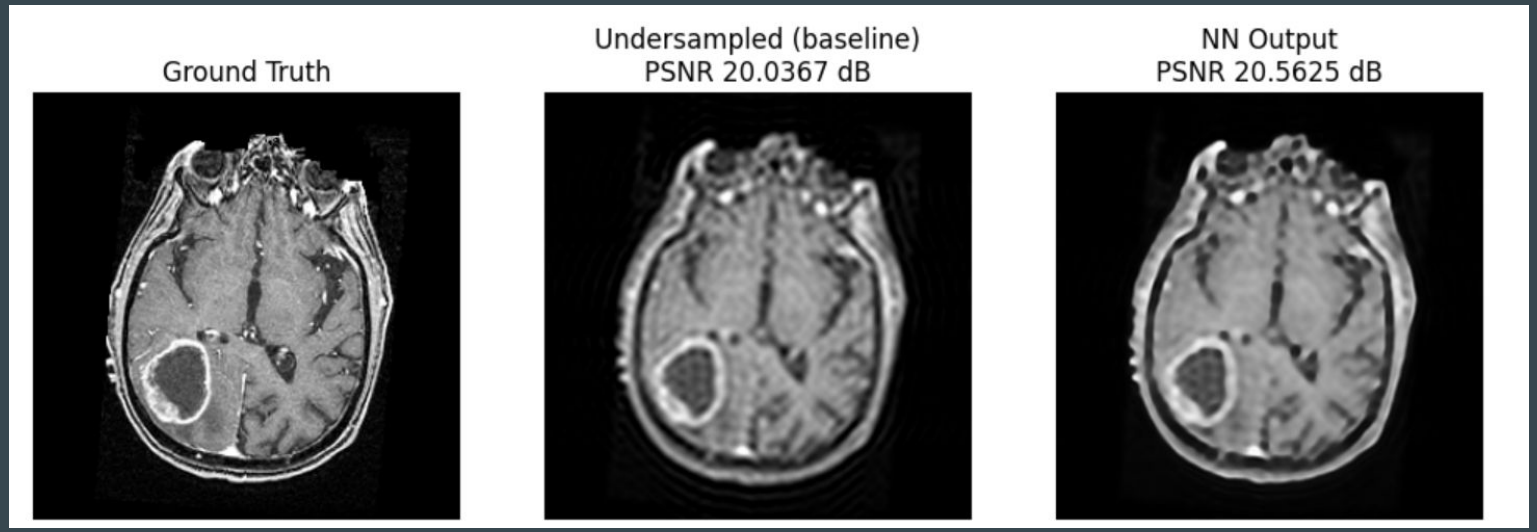
Zero-filled FFT Reconstruction



Gaussian kernel reconstruction provides highest clarity.

Initial Results pt. 2

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small residual CNN, then reconstructed the same slice we used the kernel method on



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

Next Steps

- Moving toward more general image reconstruction/upscaling due to difficulty in determining medical benchmarks
- Combining the kernel approach with neural networks
  - Use kernels to get a smooth reconstruction
  - Feed the reconstruction into a neural network
  - Network learns the residual and works as a residual corrector
  - Can only modify the unobserved regions of k-space
- Using the Structural Similarity Index as another metric of success

$$SSIM(I_{LR}, I_{HR}) = \frac{(2\mu_{I_{LR}}\mu_{I_{HR}} + c_1)(2\sigma_{I_{LR},I_{HR}} + c_2)}{(\mu_{I_{LR}}^2 + \mu_{I_{HR}}^2 + c_1)(\sigma_{I_{LR}}^2 + \sigma_{I_{HR}}^2 + c_2)}$$

- Explore different deep learning and neural network upsampling strategies (Pre, Post, and Progressive)